The Quant Grimoire

Part I - Measure Theory (based on J.F. Le Gall)

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Chapter 1

Measurable Spaces

This chapter lays the measure-theoretic foundation required to build the Itô integral. While standard texts like Le Gall focus heavily on pure mathematical rigor, our goal here is dual : to preserve that formal architecture while bridging the gap toward quantitative intuition. Before we can price a derivative or simulate a stochastic process in Python, we must first rigorously define the space of possible events and how we measure them.

> *"The starting idea of measure theory is to assign a non-negative real number (the measure of this subset) to each subset of a given set, in a way that satisfies certain natural additivity properties (the measure of a disjoint union must be the sum of the measures).*
>
> *For deep reasons, it is generally not possible to define the measure of any arbitrary subset, and one must restrict to a certain class of subsets, called* ***measurable sets****; a set equipped with a σ-algebra is called a* ***measurable space****."* — *J.-F. Le Gall [1]*

1.1 Measurable Sets and σ-Algebras

Definition 1.1: σ-Algebra

Let E be a set. A collection $\mathcal{A}$ of subsets of E is called a **σ-algebra** on E if:

(i) Certain Event : $E \in \mathcal{A}$;

(ii) Closed under Complementation : $A \in \mathcal{A} \implies A^c \in \mathcal{A}$;

(iii) Closed under Countable Unions : If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$, then $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A}$.

Integration relies on the passage to the limit. We need to sum an infinite number of small pieces to perform limit calculations. The elements of $\mathcal{A}$ are called measurable sets, or $\mathcal{A}$-measurable sets if there is any ambiguity. We say that $(E, \mathcal{A})$ is a **measurable space**.

Grimoire Insight: Intuition on σ-algebras

The definition of a σ-algebra reflects three basic physical and logical constraints on what we can measure:

- **Measuring the whole space :** We must be able to measure the total size of our universe ($\Omega \in \mathcal{A}$).
- **Measuring the rest :** If we know how to measure an event $A \in \mathcal{F}$, we must automatically know how to measure "everything else". The measure of the complement A^c is simply the measure of the total space minus that of A:

  $$\mu(A^c) = \mu(E) - \mu(A)$$

  Thus, $\mathcal{F}$ must be closed under complementation.
- **Passage to the limit :** Since analysis relies on continuous limits, if we can measure a sequence of individual events $(A_n)_{n\in\mathbb{N}}$, we must also be able to measure their combined occurrence ($\bigcup_{n=0}^{\infty} A_n \in \mathcal{A}$).

Property 1.1.1: Immediate Properties of σ-Algebras

Let $\mathcal{A}$ be a σ-algebra on a set E. The definition implies the following properties:

1. **Empty Set :** The empty set belongs to the collection :

   $$\emptyset \in \mathcal{A}$$

2. **Closure under Countable Intersections :** If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$, then their intersection is also measurable :

   $$\bigcap_{n\in\mathbb{N}} A_n \in \mathcal{A}$$

3. **Closure under Finite Operations :** By choosing $A_n = \emptyset$ (or $A_n = E$) for all n beyond a certain index, condition (iii) implies that $\mathcal{F}$ is automatically closed under **finite** unions and **finite** intersections.

Proof. We prove each assertion sequentially using axioms (i), (ii), and (iii) of a σ-algebra :

1. By axiom (i), the full space belongs to the collection ($E \in \mathcal{A}$). Applying axiom (ii), its complement must also belong to $\mathcal{A}$, so $E^c \in \mathcal{A}$. Since $E^c = \emptyset$, we directly conclude that :

   $$\emptyset \in \mathcal{A}$$

2. Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ be a sequence of measurable sets.
   - By axiom (ii), their complements are also measurable : $A_n^c \in \mathcal{A}$ for all $n \in \mathbb{N}$.
   - By axiom (iii), the countable union of these complements is measurable : $\bigcup_{n\in\mathbb{N}} A_n^c \in \mathcal{A}$.
   - Applying axiom (ii) again, the complement of this union is in $\mathcal{A}$:

     $$\left(\bigcup_{n\in\mathbb{N}} A_n^c\right)^c \in \mathcal{A}$$

By De Morgan's laws, we have $(\bigcup_{n\in\mathbb{N}} A_n^c)^c = \bigcap_{n\in\mathbb{N}} (A_n^c)^c = \bigcap_{n\in\mathbb{N}} A_n$. Thus :

$$\bigcap_{n\in\mathbb{N}} A_n \in \mathcal{A}$$

3. Let $p \in \mathbb{N}^*$ and let $A_1, \dots, A_p \in \mathcal{A}$. We construct an infinite sequence $(A_n)_{n\in\mathbb{N}}$ by padding with empty sets :

$$\forall n > p, \quad A_n = \emptyset$$

Since $\emptyset \in \mathcal{A}$, all elements of the sequence $(A_1, \dots, A_p, \emptyset, \emptyset, \dots)$ belong to $\mathcal{A}$. By axiom (iii), the countable union is in $\mathcal{A}$:

$$\bigcup_{n\in\mathbb{N}} A_n = \left(\bigcup_{i=1}^{p} A_i\right) \cup \left(\bigcup_{n>p} \emptyset\right) = \left(\bigcup_{i=1}^{p} A_i\right) \cup \emptyset = \bigcup_{i=1}^{p} A_i$$

Therefore, $\bigcup_{i=1}^{p} A_i \in \mathcal{A}$. By taking complements and using De Morgan's laws again, finite intersections $\bigcap_{i=1}^{p} A_i \in \mathcal{A}$ follow immediately.

□

Example 1.2: The Discrete (Maximal) σ-Algebra

Let E be a non-empty set. The power set $\mathcal{P}(E)$, which contains all subsets of E, forms the maximal (or discrete) σ-algebra $\mathcal{A} = \mathcal{P}(E)$ on E.

Verification. To check that $\mathcal{P}(E)$ is indeed a σ-algebra :

(i) **Entire Space :** By definition, $E \in \mathcal{P}(E)$.

(ii) **Closure under Complements :** If $A \in \mathcal{P}(E)$, then $A \subseteq E$, so $A^c = E \setminus A \subseteq E$, which implies $A^c \in \mathcal{P}(E)$.

(iii) **Closure under Countable Unions :** If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{P}(E)$, their union $\bigcup_{n\in\mathbb{N}} A_n$ remains a subset of E, hence $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{P}(E)$. □

Example 1.3: The Trivial (Minimal) σ-Algebra

Let E be a non-empty set. The collection $\mathcal{A} = \{\emptyset, E\}$, consisting only of the empty set and the entire space, forms the minimal (or trivial) σ-algebra on E.

Verification. To check that $\mathcal{A} = \{\emptyset, E\}$ satisfies the three axioms :

(i) **Entire Space :** By definition, $E \in \{\emptyset, E\}$.

(ii) **Closure under Complements :**

- If $A = \emptyset$, then $A^c = E \setminus \emptyset = E \in \{\emptyset, E\}$.
- If $A = E$, then $A^c = E \setminus E = \emptyset \in \{\emptyset, E\}$.

In both cases, $A^c \in \mathcal{A}$.

(iii) **Closure under Countable Unions :** Let $(A_n)_{n\in\mathbb{N}} \subseteq \{\emptyset, E\}$.

- If $A_n = \emptyset$ for all $n \in \mathbb{N}$, then $\bigcup_{n\in\mathbb{N}} A_n = \emptyset \in \mathcal{A}$.
- If there exists at least one n_0 such that $A_{n_0} = E$, then $\bigcup_{n\in\mathbb{N}} A_n = E \in \mathcal{A}$.

Thus, $\bigcup_{n\in\mathbb{N}} A_n \in \{\emptyset, E\}$. □

Example 1.4: The Countable / Co-Countable σ-Algebra

Let E be a non-empty set. The collection of subsets whose element or complement is at most countable:

$$\mathcal{A} = \{A \subseteq E \mid A \text{ is at most countable or } A^c \text{ is at most countable}\}$$

forms a σ-algebra on E.

Verification. We check the three axioms for $\mathcal{A}$:

(i) **Entire Space :** The complement of the whole space is $E^c = \emptyset$. Since the empty set $\emptyset$ is finite (hence at most countable), it follows that $E \in \mathcal{A}$.

(ii) **Closure under Complements :** Let $A \in \mathcal{A}$. By definition, either A or A^c is at most countable. Since $(A^c)^c = A$, the condition for A^c being in $\mathcal{A}$ is symmetric, so $A^c \in \mathcal{A}$.

(iii) **Closure under Countable Unions :** Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ and define $S = \bigcup_{n\in\mathbb{N}} A_n$.

- **Case 1 - All A_n are at most countable :** A countable union of at most countable sets is at most countable. Thus S is at most countable, so $S \in \mathcal{A}$.
- **Case 2 - At least one A_k has a countable complement :** Suppose there exists some index $k \in \mathbb{N}$ such that A_k is not at most countable. Since $A_k \in \mathcal{A}$, its complement A_k^c must be at most countable. Using De Morgan's laws:

$$S^c = \left(\bigcup_{n\in\mathbb{N}} A_n\right)^c = \bigcap_{n\in\mathbb{N}} A_n^c \subseteq A_k^c.$$

Since S^c is a subset of the at most countable set A_k^c, S^c is itself at most countable. Therefore, $S \in \mathcal{A}$.

Thus, $\mathcal{A}$ is closed under countable unions. □

Definition 1.5: Generated σ-Algebra

Let E be a non-empty set and let $\mathcal{C} \subseteq \mathcal{P}(E)$ be a collection of subsets of E. There exists a unique smallest σ-algebra on E containing $\mathcal{C}$, denoted by $\sigma(\mathcal{C})$ and called the **σ-algebra generated by** $\mathcal{C}$.
Formally, it is defined as the intersection of all σ-algebras on E that contain $\mathcal{C}$:

$$\sigma(\mathcal{C}) = \bigcap_{\substack{\mathcal{A} \text{ is a } \sigma\text{-algebra on } E \\ \mathcal{C} \subseteq \mathcal{A}}} \mathcal{A}$$

Grimoire Insight: "Forcing" the Axioms of Measure Theory

Suppose we need a σ-algebra that contains a specific collection of events $\mathcal{C}$ (e.g., asset price intervals or option payoffs). However, $\mathcal{C}$ itself rarely satisfies the three axioms of a σ-algebra.

To transform $\mathcal{C}$ into a valid σ-algebra without adding unnecessary information, we "force" the rules : we take the intersection of every existing σ-algebra that contains $\mathcal{C}$. Since the intersection of any family of σ-algebras is itself a σ-algebra, this construction yields the most minimal, economical structure containing $\mathcal{C}$.

Definition 1.6: Borel σ-Algebra

Let $(E, \mathcal{O})$ be a topological space, where $\mathcal{O}$ denotes the collection of open sets in E. The **Borel σ-algebra** on E, denoted by $\mathcal{B}(E)$, is the σ-algebra generated by all open sets of E :

$$\mathcal{B}(E) = \sigma(\mathcal{O})$$

The elements of $\mathcal{B}(E)$ are called the **Borel sets** (or **Borelians**) of E.

Grimoire Insight: Reminder - Topological Spaces

Recall that a topological space is a pair $(E, \mathcal{O})$ where E is a set and $\mathcal{O} \subseteq \mathcal{P}(E)$ is a topology (the collection of "open sets") equipping E with notions of distance, limits, and continuity. The collection $\mathcal{O}$ satisfies :

(i) **Finite Intersections :** Any finite intersection of elements in $\mathcal{O}$ belongs to $\mathcal{O}$.

(ii) **Arbitrary Unions :** Any union (finite or infinite) of elements in $\mathcal{O}$ belongs to $\mathcal{O}$.

(iii) **Extremes :** The empty set $\emptyset$ and the full space E belong to $\mathcal{O}$.

In quantitative finance and stochastic calculus, whenever E is endowed with a topology (such as $\mathbb{R}$ or $\mathbb{R}^d$ equipped with standard Euclidean metrics), it is implicitly assumed to be equipped with its Borel σ-algebra $\mathcal{B}(E)$.

Example 1.7: Borel σ-Algebra and Open Intervals

The Borel σ-algebra on $\mathbb{R}$ is generated by the collection of open intervals :

$$\mathcal{B}(\mathbb{R}) = \sigma\Big(\{]a,b[\mid \ a,b \in \mathbb{R},\, a < b\}\Big)$$

Methodology - Proof by Double Inclusion. To prove that two generated σ-algebras are equal, $\sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2)$, it suffices to show that $\mathcal{C}_1 \subseteq \sigma(\mathcal{C}_2)$ and $\mathcal{C}_2 \subseteq \sigma(\mathcal{C}_1)$. By definition of the smallest containing σ-algebra, this immediately implies $\sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2)$.

§ **First Inclusion -** $\sigma\big(\{]a,b[\}\big) \subseteq \mathcal{B}(\mathbb{R})$ **:** Every open interval $]a,b[$ is, by definition, an open set of $\mathbb{R}$. Since $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra containing all open sets of $\mathbb{R}$, it necessarily contains all open intervals :

$$\{]a,b[\mid \ a < b\} \subseteq \mathcal{B}(\mathbb{R}) \implies \sigma\Big(\{]a,b[\}\Big) \subseteq \mathcal{B}(\mathbb{R})$$

§ **Second Inclusion -** $\mathcal{B}(\mathbb{R}) \subseteq \sigma\big(\{]a,b[\}\big)$ **:** We must show that any open set of $\mathbb{R}$ can be constructed using open intervals $]a,b[$. A fundamental result in topology states that *every open set of* $\mathbb{R}$ *can be written as a countable union of disjoint open intervals*.

Since $\sigma\big(\{]a,b[\}\big)$ contains all open intervals and is closed under countable unions, it necessarily contains every open set of $\mathbb{R}$. Let $\mathcal{O}$ be the collection of open sets in $\mathbb{R}$:

$$\mathcal{O} \subseteq \sigma\Big(\{]a,b[\}\Big) \implies \mathcal{B}(\mathbb{R}) = \sigma(\mathcal{O}) \subseteq \sigma\Big(\{]a,b[\}\Big)$$

Combining both inclusions yields $\mathcal{B}(\mathbb{R}) = \sigma\big(\{]a,b[\}\big)$. □

Example 1.8: Equivalence of Generators: $]a,b[$ vs $]-\infty,a]$

We have the equivalence between the σ-algebras generated by bounded open intervals and left-closed ray intervals :

$$\sigma\Big(\{]a,b[\mid \ a < b\}\Big) = \sigma\Big(\{]-\infty,a] \ \mid \ a \in \mathbb{R}\}\Big)$$

Methodology. We establish both inclusions by constructing each set type using countable operations :

§ **First Step - Constructing** $]-\infty,a]$ **from** $]a,b[$ **:**

- We first construct the open ray $]-\infty,a[$ as a countable union of open intervals :

$$]-\infty,a[= \bigcup_{n=1}^{\infty}]-n,a[$$

This countable union of elements in $\sigma\big(\{]a,b[\}\big)$ belongs to $\sigma\big(\{]a,b[\}\big)$.

- Next, we isolate the singleton $\{a\}$ as a countable intersection of open intervals :

$$\{a\} \;=\; \bigcap_{n=1}^{\infty} \Big]a - \frac{1}{n}, a + \frac{1}{n}\Big[$$

Thus, $\{a\} \in \sigma\big(\{]a, b[\}\big)$.

Combining these two sets via a finite union :

$$]-\infty, a] \;=]-\infty, a[\cup \{a\}$$

Since both sets are in $\sigma\big(\{]a, b[\}\big)$, their union is as well. This proves :

$$\sigma\Big(\big\{]-\infty, a]\big\}\Big) \subseteq \sigma\Big(\big\{]a, b[\big\}\Big)$$

§ **Second Step - Constructing $]a, b[$ from $]-\infty, a]$:**

- Note that the left-semi-open interval $]a, b]$ can be expressed as a difference of two rays :

$$]a, b] \;=]-\infty, b] \setminus]-\infty, a] \;=]-\infty, b] \;\cap\; \big(]-\infty, a]\big)^c$$

Since $\sigma\big(\{]-\infty, a]\}\big)$ is closed under complements and intersections, $]a, b] \in \sigma\big(\{]-\infty, a]\}\big)$.

- Now, we express the open interval $]a, b[$ as a countable union of semi-open intervals :

$$]a, b[= \bigcup_{n=1}^{\infty} \Big]a, b - \frac{1}{n}\Big]$$

Since this is a countable union of elements in $\sigma\big(\{]-\infty, a]\}\big)$, we obtain $]a, b[\in \sigma\big(\{]-\infty, a]\}\big)$. This proves :

$$\sigma\Big(\big\{]a, b[\big\}\Big) \subseteq \sigma\Big(\big\{]-\infty, a]\big\}\Big)$$

Both building blocks have been successfully constructed with basic set operations and countable limits, completing the proof of equality. □

Definition 1.9: Product σ-Algebra

Let $(E_1, \mathcal{A}_1)$ and $(E_2, \mathcal{A}_2)$ be two measurable spaces. The **product σ-algebra** on the Cartesian product space $E_1 \times E_2$, denoted by $\mathcal{A}_1 \otimes \mathcal{A}_2$, is the σ-algebra generated by all measurable rectangles :

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma\Big(\big\{A_1 \times A_2 \;\mid\; A_1 \in \mathcal{A}_1,\, A_2 \in \mathcal{A}_2\big\}\Big)$$

Grimoire Insight: Intuition - Measuring in Higher Dimensions

Suppose we know how to measure events in E_1 (e.g., stock prices) and in E_2 (e.g., interest rates). To build a unified model on the product space $E_1 \times E_2$ (such as the 2D plane $\mathbb{R}^2$), the simplest geometric building block is a "rectangle" of the form $A_1 \times A_2$.
Crucial Warning : The set of all measurable rectangles $\{A_1 \times A_2 \mid A_1 \in \mathcal{A}_1, A_2 \in \mathcal{A}_2\}$ does *not* form a σ-algebra on its own!

- **Visual Proof - Unions :** The union of two crossed rectangles in the plane forms a cross-shaped region (+), which is clearly not a single rectangle.
- **Visual Proof - Complements :** The complement of a rectangle in the plane is not a rectangle either.

Since rectangles are not closed under basic set operations, they fail to form a σ-algebra. We must therefore **generate** the σ-algebra by taking the smallest σ-algebra containing all such measurable rectangles, leading directly to the definition $\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma(\{A_1 \times A_2\})$.

Theorem 1.10: Borel Measurability Theorem

The preimage of a Borel set under a continuous map is a Borel set.

Proof. Let $f \in \mathcal{C}^0(E, F)$ and $\mathcal{M} = \{B \in \mathcal{B}(F) \mid f^{-1}(B) \in \mathcal{B}(E)\}$ be the collection of subsets of F whose preimage under f is measurable in E.

(i) **Open sets containment :** Let $O \in \mathcal{O}_F$ be an open set in F. By continuity of f, the preimage $f^{-1}(O)$ is open in E. Since $\mathcal{B}(E)$ contains all open sets of E, we have $f^{-1}(O) \in \mathcal{B}(E)$, which means $\mathcal{O}_F \subseteq \mathcal{M}$.

(ii) **σ-Algebra structure :** The inverse image operation f^{-1} preserves complements and countable unions. Thus, $\mathcal{M}$ forms a σ-algebra on F.

Since $\mathcal{M}$ is a σ-algebra containing all open sets $\mathcal{O}_F$, it necessarily contains the smallest σ-algebra generated by them, i.e., $\mathcal{B}(F) = \sigma(\mathcal{O}_F) \subseteq \mathcal{M}$. Hence, $\mathcal{M} = \mathcal{B}(F)$, proving that the preimage of every Borel set is a Borel set. □

Example 1.11: Borel σ-Algebra on $\mathbb{R}^2$

The 2D Borel σ-algebra on $\mathbb{R}^2$ coincides with the product of 1D Borel σ-algebras :

$$\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$$

Methodology - Proof by Double Inclusion. We establish equality by proving both inclusions :

§ **First Inclusion -** $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$ **:** Let $A, B \in \mathcal{B}(\mathbb{R})$. The rectangle $A \times B \subseteq \mathbb{R}^2$ can be expressed as the intersection of two infinite strips :

$$A \times B = (A \times \mathbb{R}) \cap (\mathbb{R} \times B)$$

Consider the canonical projection maps :

$$\pi_1 : \mathbb{R}^2 \to \mathbb{R}, \quad (x, y) \mapsto x, \qquad \text{and} \qquad \pi_2 : \mathbb{R}^2 \to \mathbb{R}, \quad (x, y) \mapsto y$$

Both projections are continuous. By the Borel measurability theorem above, π_1 and π_2 are $(\mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}))$-measurable. Thus :

$$A \times \mathbb{R} = \pi_1^{-1}(A) \in \mathcal{B}(\mathbb{R}^2) \qquad \text{and} \qquad \mathbb{R} \times B = \pi_2^{-1}(B) \in \mathcal{B}(\mathbb{R}^2)$$

Since $\mathcal{B}(\mathbb{R}^2)$ is closed under finite intersections, the rectangle $A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B)$ belongs to $\mathcal{B}(\mathbb{R}^2)$. As $\mathcal{B}(\mathbb{R}^2)$ contains all measurable rectangles, it contains the generated product σ-algebra :

$$\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$$

§ **Second Inclusion -** $\mathcal{B}(\mathbb{R}^2) \subseteq \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$ **:** Recall that $\mathcal{B}(\mathbb{R}^2)$ is generated by the open sets of $\mathbb{R}^2$. It suffices to show that any open set $U \subseteq \mathbb{R}^2$ belongs to $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$.

A standard topological result states that every open set $U \subseteq \mathbb{R}^2$ can be written as an at most countable union of open rectangles with rational endpoints:

$$U = \bigcup_{n \in \mathbb{N}} \left( ]q_{1,n}, q_{2,n}[\times]q_{3,n}, q_{4,n}[\right), \qquad \text{with } q_{i,n} \in \mathbb{Q}$$

Each open interval $]q_1, q_2[$ is in $\mathcal{B}(\mathbb{R})$, so each open rectangle belongs to the product generator set. Since $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$ is closed under countable unions, it contains U. Thus :

$$\mathcal{B}(\mathbb{R}^2) \subseteq \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$$

Combining both inclusions completes the proof : $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. □

1.2 Positive Measures

Definition 1.12: Positive Measure

Let $(E, \mathcal{A})$ be a measurable space. A **positive measure** on $(E, \mathcal{A})$ is a function :

$$\mu : \mathcal{A} \to [0, +\infty]$$

satisfying the following two properties :

(i) **Null Empty Set :** The measure of the empty set is zero :

$$\mu(\emptyset) = 0$$

(ii) **σ-Additivity :** For any pairwise disjoint family of measurable sets $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ (i.e. $A_i \cap A_j = \emptyset$ for all $i \neq j$) :

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n=0}^{\infty} \mu(A_n)$$

Grimoire Insight: Intuition - Scale and Area Decomposition

In plain terms, a measure acts simply as a "scale" or a "graduated ruler".

- Measuring nothing yields nothing.
- Furthermore, if a piece of land is partitioned into a countable infinity of strictly disjoint sub-plots, the total area of the land equals the infinite sum of the individual sub-plot areas.

Property 1.2.1: Fundamental Properties of Positive Measures

Let $(E, \mathcal{A})$ be a measurable space and let $\mu : \mathcal{A} \to [0, +\infty]$ be a positive measure. The following properties hold :

1. **Monotonicity and Subtraction :** If $A, B \in \mathcal{A}$ with $A \subseteq B$, then :

   $$\mu(A) \leq \mu(B)$$

   If, in addition, $\mu(A) < +\infty$, then :

   $$\mu(B \setminus A) = \mu(B) - \mu(A)$$

2. **Inclusion-Exclusion Principle (Poincaré's Formula) :** For any $A, B \in \mathcal{A}$:

   $$\mu(A) + \mu(B) = \mu(A \cup B) + \mu(A \cap B)$$

3. **Continuity from Below (Increasing Sequences) :** If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ is an increasing sequence of measurable sets ($A_n \subseteq A_{n+1}$ for all n), then :

   $$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \lim_{n\to\infty} \mu(A_n)$$

4. **Continuity from Above (Decreasing Sequences) :** If $(B_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ is a decreasing sequence of measurable sets ($B_{n+1} \subseteq B_n$ for all n) and $\mu(B_0) < +\infty$, then :

$$\mu\left(\bigcap_{n\in\mathbb{N}} B_n\right) = \lim_{n\to\infty} \mu(B_n)$$

5. **Countable Sub-additivity :** For any sequence of measurable sets $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ (not necessarily disjoint) :

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n=0}^{\infty} \mu(A_n)$$

Proof. We establish each property using the axioms of a positive measure :

1. **Monotonicity and Subtraction :** Since $A \subseteq B$, we decompose B into a disjoint union : $B = A \cup (B \setminus A)$. By finite additivity :

$$\mu(B) = \mu(A) + \mu(B \setminus A)$$

Since μ takes values in $[0, +\infty]$, $\mu(B \setminus A) \geq 0$, which yields $\mu(B) \geq \mu(A)$. If $\mu(A) < +\infty$, subtracting $\mu(A)$ from both sides gives $\mu(B \setminus A) = \mu(B) - \mu(A)$.

2. **Inclusion-Exclusion Formula :** We express $A \cup B$ and B using disjoint unions :

$$A \cup B = A \cup (B \setminus A) \implies \mu(A \cup B) = \mu(A) + \mu(B \setminus A)$$

$$B = (A \cap B) \cup (B \setminus A) \implies \mu(B) = \mu(A \cap B) + \mu(B \setminus A)$$

Substituting $\mu(B \setminus A) = \mu(B) - \mu(A \cap B)$ into the first identity yields :

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B) \implies \mu(A) + \mu(B) = \mu(A \cup B) + \mu(A \cap B)$$

3. **Continuity from Below :** To apply σ-additivity, we construct a pairwise disjoint sequence $(C_n)_{n\in\mathbb{N}}$

$$C_0 = A_0, \qquad \forall n \geq 1, \quad C_n = A_n \setminus A_{n-1}$$

By construction, the C_n are disjoint and $\bigcup_{n\in\mathbb{N}} A_n = \bigcup_{n\in\mathbb{N}} C_n$, with $\bigcup_{k=0}^{n} C_k = A_n$ for every n. By σ-additivity :

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \mu\left(\bigcup_{n\in\mathbb{N}} C_n\right) = \sum_{n=0}^{\infty} \mu(C_n) = \lim_{N\to\infty} \sum_{n=0}^{N} \mu(C_n) = \lim_{N\to\infty} \mu(A_N)$$

4. **Continuity from Above :** Define $A_n = B_0 \setminus B_n$ for all $n \in \mathbb{N}$. Since $(B_n)_{n\in\mathbb{N}}$ is decreasing, $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of measurable sets with $\bigcup_{n\in\mathbb{N}} A_n = B_0 \setminus (\bigcap_{n\in\mathbb{N}} B_n)$. Applying continuity from below to $(A_n)_{n\in\mathbb{N}}$:

$$\mu\left(B_0 \setminus \bigcap_{n\in\mathbb{N}} B_n\right) = \lim_{n\to\infty} \mu(A_n) = \lim_{n\to\infty} \mu(B_0 \setminus B_n)$$

Since $\mu(B_0) < +\infty$, we use property (i) to write $\mu(B_0 \setminus B_n) = \mu(B_0) - \mu(B_n)$ and $\mu\left(B_0 \setminus \bigcap B_n\right) = \mu(B_0) - \mu\left(\bigcap B_n\right)$. Subtracting $\mu(B_0)$ gives :

$$\mu\left(\bigcap_{n\in\mathbb{N}} B_n\right) = \lim_{n\to\infty} \mu(B_n)$$

5. **Countable Sub-additivity :** We make the sequence disjoint by setting $C_0 = A_0$ and $C_n = A_n \setminus \left(\bigcup_{k=0}^{n-1} A_k\right)$ for $n \geq 1$. The $(C_n)_{n\in\mathbb{N}}$ are disjoint, $C_n \subseteq A_n$ for all n, and $\bigcup_{n\in\mathbb{N}} A_n = \bigcup_{n\in\mathbb{N}} C_n$. By σ-additivity and monotonicity :

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \mu\left(\bigcup_{n\in\mathbb{N}} C_n\right) = \sum_{n=0}^{\infty} \mu(C_n) \leq \sum_{n=0}^{\infty} \mu(A_n)$$

□

Example 1.13: The Counting Measure

Let E be a non-empty set and let $\mathcal{A} = \mathcal{P}(E)$. The **counting measure** is the function $\mu : \mathcal{P}(E) \to [0, +\infty]$ defined for any subset $A \subseteq E$ by :

$$\mu(A) = \text{Card}(A) = \begin{cases} n & \text{if } A \text{ is a finite set with } n \text{ elements} \\ +\infty & \text{if } A \text{ is infinite} \end{cases}$$

Verification. We verify the two axioms of a positive measure :

(i) **Null Empty Set :** By definition of cardinality, $\mu(\emptyset) = \text{Card}(\emptyset) = 0$.

(ii) **σ-Additivity :** Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{P}(E)$ be a sequence of pairwise disjoint subsets.

- If at least one of the A_n is infinite, then the union $\bigcup_{n\in\mathbb{N}} A_n$ is also infinite. Thus :

$$\text{Card}\left(\bigcup_{n\in\mathbb{N}} A_n\right) = +\infty = \sum_{n=0}^{\infty} \text{Card}(A_n)$$

- If infinitely many A_n are non-empty, the union contains infinitely many disjoint elements, so :

$$\text{Card}\left(\bigcup_{n\in\mathbb{N}} A_n\right) = +\infty = \sum_{n=0}^{\infty} \text{Card}(A_n)$$

- If only a finite number k of the A_n are non-empty and all are finite, the property follows directly from the finite additivity of cardinality for disjoint sets :

$$\text{Card}\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n=0}^{k} \text{Card}(A_n) = \sum_{n=0}^{\infty} \text{Card}(A_n)$$

Hence, μ is σ-additive.

□

Example 1.14: The Dirac Measure and Linear Combinations

Let $(E, \mathcal{A})$ be a measurable space and let $x \in E$ be a fixed point. The **Dirac measure** centered at x, denoted by δ_x, is defined for any $A \in \mathcal{A}$ by :

$$\delta_x(A) = \mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

More generally, given a sequence of points $(x_n)_{n\in\mathbb{N}} \subseteq E$ and non-negative coefficients $(\alpha_n)_{n\in\mathbb{N}} \subseteq [0, +\infty]$, the map :

$$\mu(A) = \sum_{n=0}^{\infty} \alpha_n \delta_{x_n}(A) = \sum_{n=0}^{\infty} \alpha_n \mathbf{1}_A(x_n)$$

defines a valid positive measure on $(E, \mathcal{A})$ via stability under positive linear combinations and pointwise limits.

Verification for the Dirac Measure. We check the two axioms for δ_x :

(i) **Null Empty Set:** Since $x \notin \emptyset$, we have $\delta_x(\emptyset) = 0$.

(ii) σ**-Additivity:** Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ be a sequence of pairwise disjoint measurable sets.

- If $x \in \bigcup_{n\in\mathbb{N}} A_n$, since the sets are disjoint, there exists a unique index $k \in \mathbb{N}$ such that $x \in A_k$. Thus :

$$\delta_x\left(\bigcup_{n\in\mathbb{N}} A_n\right) = 1 \qquad \text{and} \qquad \sum_{n=0}^{\infty} \delta_x(A_n) = \delta_x(A_k) = 1$$

- If $x \notin \bigcup_{n\in\mathbb{N}} A_n$, then $x \notin A_n$ for all $n \in \mathbb{N}$. Both sides evaluate to 0 :

$$\delta_x\left(\bigcup_{n\in\mathbb{N}} A_n\right) = 0 = \sum_{n=0}^{\infty} \delta_x(A_n)$$

Thus, δ_x is a positive measure.

□

Example 1.15: The Lebesgue Measure on $\mathbb{R}$ (Admitted for now)

There exists a unique positive measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, denoted by λ, such that for every open interval $]a, b[$ with $a \leq b$:

$$\lambda(]a, b[) = b - a$$

The Lebesgue measure λ provides the rigorous mathematical formalization of the intuitive notion of "length" in $\mathbb{R}$.

Definition 1.16: Finite Measure

A positive measure μ on $(E, \mathcal{A})$ is said to be **finite** if the total mass of the space is finite :

$$\mu(E) < +\infty$$

Definition 1.17: Probability Measure

A positive measure μ on $(E, \mathcal{A})$ is called a **probability measure** if the total mass of the space equals one :

$$\mu(E) = 1$$

In probability theory, the state space E is standardly denoted by Ω, the σ-algebra $\mathcal{A}$ by $\mathcal{F}$, and the probability measure μ by $\mathbb{P}$.

Definition 1.18: σ-Finite Measure

A positive measure μ on $(E, \mathcal{A})$ is said to be **σ-finite** if there exists an increasing sequence of measurable sets $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ such that :

$$E = \bigcup_{n\in\mathbb{N}} E_n \qquad \text{and} \qquad \forall n \in \mathbb{N}, \quad \mu(E_n) < +\infty$$

Grimoire Insight: Intuition - σ-Finite vs. Finite

In plain terms, a space is σ-finite if it can be decomposed into countably many pieces of finite size. Note that σ-finite **does not imply** finite (σ-finite $\neq$ finite).

Classic Example : The Lebesgue measure λ on $\mathbb{R}$ is not finite since $\lambda(\mathbb{R}) = +\infty$. However, setting $E_n = [-n, n]$ gives an increasing sequence of measurable sets such that $\bigcup_{n\in\mathbb{N}} E_n = \mathbb{R}$ and $\lambda(E_n) = 2n < +\infty$ for all $n \in \mathbb{N}$. Thus, λ is σ-finite.

Definition 1.19: Atom

Let $(E, \mathcal{A})$ be a measurable space equipped with a positive measure μ such that singletons $\{x\} \in \mathcal{A}$ for all $x \in E$. A point $x \in E$ is called an **atom** of μ if its individual measure is strictly positive :

$$\mu(\{x\}) > 0$$

Definition 1.20: Diffuse Measure

A positive measure μ on $(E, \mathcal{A})$ is said to be **diffuse** (or non-atomic) if it possesses no atoms :

$$\forall x \in E, \quad \mu(\{x\}) = 0$$

1.3 Measurable Functions

Definition 1.21: Measurable Function

Let $(E, \mathcal{A})$ and $(F, \mathcal{B})$ be two measurable spaces. A function $f : E \to F$ is said to be $(\mathcal{A}, \mathcal{B})$**-measurable** (or simply **measurable**) if the preimage of every measurable set in F is measurable in E:

$$\forall B \in \mathcal{B}, \quad f^{-1}(B) \in \mathcal{A}$$

When E and F are topological spaces equipped with their respective Borel σ-algebras $\mathcal{B}(E)$ and $\mathcal{B}(F)$, a $(\mathcal{B}(E), \mathcal{B}(F))$-measurable function is also called a **Borel function** (or **Borelian**).

Proposition 1.22: Stability under Composition

The composition of two measurable functions is measurable.

Proof. Let $(E, \mathcal{A})$, $(F, \mathcal{B})$, and $(G, \mathcal{C})$ be three measurable spaces. Let $f : E \to F$ be $(\mathcal{A}, \mathcal{B})$-measurable and $g : F \to G$ be $(\mathcal{B}, \mathcal{C})$-measurable.

Let $N \in \mathcal{C}$. By standard properties of preimages :

$$(g \circ f)^{-1}(N) = f^{-1}\big(g^{-1}(N)\big)$$

Since g is measurable, $M = g^{-1}(N) \in \mathcal{B}$. Furthermore, since f is measurable and $M \in \mathcal{B}$, we have $f^{-1}(M) \in \mathcal{A}$. Thus, $(g \circ f)^{-1}(N) = f^{-1}(M) \in \mathcal{A}$, which proves that $g \circ f$ is $(\mathcal{A}, \mathcal{C})$-measurable. □

Proposition 1.23: Sufficient Condition for Measurability

Let $(E, \mathcal{A})$ and $(F, \mathcal{B})$ be two measurable spaces. For a function $f : E \to F$ to be measurable, it suffices that there exists a generating collection $\mathcal{C} \subseteq \mathcal{B}$ such that $\sigma(\mathcal{C}) = \mathcal{B}$ and :

$$\forall B \in \mathcal{C}, \quad f^{-1}(B) \in \mathcal{A}$$

Proof. Define the collection of sets in F whose preimage under f is measurable in E:

$$\mathcal{G} = \{B \in \mathcal{B} \mid f^{-1}(B) \in \mathcal{A}\}$$

It is straightforward to verify that $\mathcal{G}$ forms a σ-algebra on F (since the preimage operation preserves complements and countable unions).

By assumption, $\mathcal{C} \subseteq \mathcal{G}$. Since $\mathcal{G}$ is a σ-algebra containing $\mathcal{C}$, it must contain the smallest σ-algebra generated by $\mathcal{C}$, which means:

$$\mathcal{B} = \sigma(\mathcal{C}) \subseteq \mathcal{G}$$

Since $\mathcal{G} \subseteq \mathcal{B}$ by construction, we obtain $\mathcal{G} = \mathcal{B}$. Therefore, for every $B \in \mathcal{B}$, $f^{-1}(B) \in \mathcal{A}$, establishing that f is measurable. □

Example 1.24: Measurability of Real-Valued Functions

Let $(E, \mathcal{A})$ be a measurable space and let $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be the real line equipped with its Borel σ-algebra. A function $f : E \to \mathbb{R}$ is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$-measurable if and only if for every open interval $]a, b[$ (with $a, b \in \mathbb{R}, a < b$) :

$$f^{-1}(]a, b[) \in \mathcal{A}$$

Proof. Recall that $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}_1)$ where $\mathcal{C}_1 = \{]a, b[|\ a, b \in \mathbb{R}, a < b\}$. Using the sufficient condition for measurability established in the previous proposition, it suffices to verify the preimage condition on the generating collection $\mathcal{C}_1$.

Since $f^{-1}(]a, b[) \in \mathcal{A}$ for every generator $]a, b[\in \mathcal{C}_1$, it follows directly that f is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$-measurable. □

Example 1.25: Continuous Functions are Borelian

Let $(E, \mathcal{O}_E)$ and $(F, \mathcal{O}_F)$ be two topological spaces equipped with their Borel σ-algebras $\mathcal{B}(E) = \sigma(\mathcal{O}_E)$ and $\mathcal{B}(F) = \sigma(\mathcal{O}_F)$.
Any continuous function $f : E \to F$ is $(\mathcal{B}(E), \mathcal{B}(F))$ - measurable.

Proof. Let $\mathcal{C} = \mathcal{O}_F$ be the collection of open sets in F, which generates $\mathcal{B}(F)$, i.e. $\sigma(\mathcal{C}) = \mathcal{B}(F)$.

Let $V \in \mathcal{C}$ be an open set in F. Since f is continuous, the preimage $f^{-1}(V)$ is an open set in E, which means :

$$f^{-1}(V) \in \mathcal{O}_E \subseteq \mathcal{B}(E)$$

Thus, for every $V \in \mathcal{C}$, we have $f^{-1}(V) \in \mathcal{B}(E)$.

By the sufficient condition for measurability applied to the generating class $\mathcal{C} = \mathcal{O}_F$, we conclude that f is $(\mathcal{B}(E), \mathcal{B}(F))$-measurable. □

Grimoire Insight: Intuition

Intuitively, since the preimage of an open set under a continuous function is open, and the Borel σ-algebra is generated by open sets, measurability follows immediately.

Lemma 1.26: Measurability of Product Maps

Let $(E, \mathcal{A})$, $(F_1, \mathcal{B}_1)$, and $(F_2, \mathcal{B}_2)$ be three measurable spaces. Let $f_1 : E \to F_1$ be $(\mathcal{A}, \mathcal{B}_1)$ - measurable and $f_2 : E \to F_2$ be $(\mathcal{A}, \mathcal{B}_2)$ - measurable.
Then the product map $f : E \to F_1 \times F_2$, $x \mapsto (f_1(x), f_2(x))$, is $(\mathcal{A}, \mathcal{B}_1 \otimes \mathcal{B}_2)$ - measurable.

Proof. By definition, the product σ-algebra $\mathcal{B}_1 \otimes \mathcal{B}_2$ is generated by the collection of measurable rectangles :

$$\mathcal{B}_1 \otimes \mathcal{B}_2 = \sigma(\mathcal{C}) \qquad \text{where} \qquad \mathcal{C} = \{B_1 \times B_2 \mid B_1 \in \mathcal{B}_1,\ B_2 \in \mathcal{B}_2\}$$

By the sufficient condition for measurability, it suffices to verify that the preimage of every generator $B_1 \times B_2 \in \mathcal{C}$ belongs to the σ-algebra $\mathcal{A}$.

Let $B_1 \in \mathcal{B}_1$ and $B_2 \in \mathcal{B}_2$. We decompose the preimage set $f^{-1}(B_1 \times B_2)$:

$$x \in f^{-1}(B_1 \times B_2) \iff f(x) \in B_1 \times B_2 \iff \big(f_1(x), f_2(x)\big) \in B_1 \times B_2$$

$$\iff f_1(x) \in B_1 \text{ and } f_2(x) \in B_2 \iff x \in f_1^{-1}(B_1) \cap f_2^{-1}(B_2)$$

This establishes the fundamental identity :

$$f^{-1}(B_1 \times B_2) = f_1^{-1}(B_1) \cap f_2^{-1}(B_2)$$

Since f_1 and f_2 are measurable, $f_1^{-1}(B_1) \in \mathcal{A}$ and $f_2^{-1}(B_2) \in \mathcal{A}$. A σ-algebra being closed under finite intersections, we obtain $f^{-1}(B_1 \times B_2) \in \mathcal{A}$, proving that f is $(\mathcal{A}, \mathcal{B}_1 \otimes \mathcal{B}_2)$ - measurable. □

Proposition 1.27: Converse of the Product Map Lemma

Let $f = (f_1, f_2) : E \to F_1 \times F_2$ be $(\mathcal{A}, \mathcal{B}_1 \otimes \mathcal{B}_2)$-measurable. Then f_1 is $(\mathcal{A}, \mathcal{B}_1)$ - measurable and f_2 is $(\mathcal{A}, \mathcal{B}_2)$ - measurable.

Proof. Let $B_1 \in \mathcal{B}_1$. Consider the particular measurable rectangle $B_1 \times F_2 \in \mathcal{B}_1 \otimes \mathcal{B}_2$. Since f is measurable, its preimage belongs to $\mathcal{A}$:

$$f^{-1}(B_1 \times F_2) = f_1^{-1}(B_1) \cap f_2^{-1}(F_2) = f_1^{-1}(B_1) \cap E = f_1^{-1}(B_1) \in \mathcal{A}$$

which proves that f_1 is $(\mathcal{A}, \mathcal{B}_1)$ - measurable. By symmetry, choosing the rectangle $F_1 \times B_2$ with $B_2 \in \mathcal{B}_2$ shows that $f_2^{-1}(B_2) \in \mathcal{A}$, establishing that f_2 is $(\mathcal{A}, \mathcal{B}_2)$ - measurable. □

We obtain a double implication, so an equivalence, written in the following theorem.

Theorem 1.28: Characterization of Measurable Product Maps

Let $(E, \mathcal{A})$, $(F_1, \mathcal{B}_1)$, and $(F_2, \mathcal{B}_2)$ be three measurable spaces. Let $f_1 : E \to F_1$ and $f_2 : E \to F_2$ be two functions, and consider the product map $f : E \to F_1 \times F_2$ given by $x \mapsto (f_1(x), f_2(x))$.

Then, f is $(\mathcal{A}, \mathcal{B}_1 \otimes \mathcal{B}_2)$ - measurable if and only if f_1 is $(\mathcal{A}, \mathcal{B}_1)$ - measurable and f_2 is $(\mathcal{A}, \mathcal{B}_2)$ - measurable :

f is $(\mathcal{A}, \mathcal{B}_1 \otimes \mathcal{B}_2)$-measurable $\iff$ f_1 is $(\mathcal{A}, \mathcal{B}_1)$-measurable and f_2 is $(\mathcal{A}, \mathcal{B}_2)$-measurable

Corollary 1.3.1: Stability under Algebraic Operations

Let $(E, \mathcal{A})$ be a measurable space and let $f, g : (E, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be two measurable real-valued functions. Then :

1. $f + g$ and $f \cdot g$ are measurable.
2. $\inf(f, g)$ and $\sup(f, g)$ are measurable.
3. The positive part $f^+ = \sup(f, 0)$ and the negative part $f^- = \sup(-f, 0)$ are measurable.

Proof. We establish each part by decomposing the operations into compositions of product maps and continuous functions :

1. Consider the sum $f + g$. We decompose it as the composition of two functions :

$$E \xrightarrow{H} \mathbb{R}^2 \xrightarrow{\Phi} \mathbb{R}$$

where :

$$H : E \to \mathbb{R}^2, \quad x \mapsto \big(f(x), g(x)\big)$$
$$\Phi : \mathbb{R}^2 \to \mathbb{R}, \quad (a, b) \mapsto a + b$$

- Since f and g are measurable, the characterization theorem for product maps implies that H is $(\mathcal{A}, \mathcal{B}(\mathbb{R}^2))$ - measurable.
- Furthermore, Φ is a continuous linear map between finite-dimensional normed vector spaces, hence it is continuous and therefore $(\mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}))$-measurable.

Since the composition of two measurable functions is measurable, $f + g = \Phi \circ H$ is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$ - measurable.

2. The exact same compositional reasoning applies by replacing $\Phi(a, b)$ with the appropriate continuous function :

- For $f \cdot g$, set $\Phi(a, b) = a \cdot b$, which is continuous on $\mathbb{R}^2$.
- For $\inf(f, g)$, set $\Phi(a, b) = \min(a, b)$, which is continuous on $\mathbb{R}^2$.
- For $\sup(f, g)$, set $\Phi(a, b) = \max(a, b)$, which is continuous on $\mathbb{R}^2$.

In all three cases, Φ is continuous (hence Borelian), so the composition $\Phi \circ H$ yields the desired measurability.

3. The positive and negative parts are immediate special cases of the supremum by taking $g = 0$ (the constant zero function, which is trivial measurable) :

$$f^+ = \sup(f, 0) \qquad \text{and} \qquad f^- = \sup(-f, 0)$$

Applying part (2) directly yields the measurability of both f^+ and f^-.

□

Grimoire Insight: Classic Function Decompositions

Any real-valued function f can be decomposed into its non-negative components :

$$f = f^+ - f^- \qquad \text{and} \qquad |f| = f^+ + f^-$$

Since f^+ and f^- are measurable, this classical decomposition immediately implies that if f is measurable, then its absolute value $|f|$ is also measurable.

Grimoire Insight: Recall - Limit Superior and Limit Inferior

For a real sequence $(a_n)_{n\in\mathbb{N}}$, the standard limit $\lim_{n\to\infty} a_n$ does not always exist (e.g. $a_n = (-1)^n$). However, in the extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, the limit superior (lim sup) and limit inferior (lim inf) always exist.

- The quantity $\sup_{k\geq n} a_k$ represents the "worst-case scenario starting from index n". As n increases, the supremum is taken over a smaller set, so the sequence $u_n = \sup_{k\geq n} a_k$ is non-increasing. It necessarily admits a limit in $\overline{\mathbb{R}}$:

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \downarrow \left(\sup_{k\geq n} a_k \right)$$

- Similarly, $v_n = \inf_{k\geq n} a_k$ forms a non-decreasing sequence, giving :

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \uparrow \left(\inf_{k\geq n} a_k \right)$$

Example 1.29: Study of lim sup and lim inf for a Sequence of Functions

Consider the sequence of functions $(f_n)_{n\in\mathbb{N}^*}$ defined on $E = [0,1]$ with values in $\mathbb{R}$ by :

$$f_n : [0,1] \to \mathbb{R}, \quad x \mapsto \begin{cases} x^n & \text{if } n \text{ is even} \\ x & \text{if } n \text{ is odd} \end{cases}$$

Study the measurability of each f_n, as well as that of $\limsup_{n\to\infty} f_n$ and $\liminf_{n\to\infty} f_n$.

Proof.

1. **Measurability of f_n :** For each $n \in \mathbb{N}^*$, f_n is continuous on $E = [0,1]$ as it consists of restriction of polynomial functions. Since any continuous function is Borelian, each f_n is $(\mathcal{B}([0,1]), \mathcal{B}(\mathbb{R}))$-measurable.

2. **Pointwise Study of $\limsup f_n(x)$ and $\liminf f_n(x)$:** Fix $x \in [0,1]$ and analyze the real sequence $\left(f_n(x)\right)_{n\in\mathbb{N}^*}$:

- **Case 1 ($x = 0$) :** For all $n \geq 1$, $f_n(0) = 0$. Hence :

$$\limsup_{n\to\infty} f_n(0) = 0 = \liminf_{n\to\infty} f_n(0)$$

- **Case 2 ($x = 1$) :** For all $n \geq 1$, $f_n(1) = 1$. Hence :

$$\limsup_{n\to\infty} f_n(1) = 1 = \liminf_{n\to\infty} f_n(1)$$

- **Case 3 ($x \in]0,1[$) :** The sequence alternates between two sub-sequences :

$$f_{2k}(x) = x^{2k} \xrightarrow{k\to\infty} 0$$

$$f_{2k+1}(x) = x \quad \text{(constant strictly positive value)}$$

Taking the limits of the supremum and infimum over sub-sequences yields :

$$\limsup_{n\to\infty} f_n(x) = x \qquad \text{and} \qquad \liminf_{n\to\infty} f_n(x) = 0$$

3. **Conclusion and Measurability of Limit Functions :** Synthesizing the results for all $x \in E$:

- For the limit superior :

$$\forall x \in E, \quad \limsup_{n\to\infty} f_n(x) = \mathrm{id}_E(x) = x$$

The identity map is continuous, hence $\limsup_{n\to\infty} f_n$ is measurable.

- For the limit inferior :

$$\forall x \in E, \quad \liminf_{n\to\infty} f_n(x) = \begin{cases} 1 & \text{if } x = 1 \\ 0 & \text{if } x \in [0,1[\end{cases} = \mathbf{1}_{\{1\}}(x)$$

Since $\{1\}$ is a closed Borel set in $[0,1]$, the indicator function $\mathbf{1}_{\{1\}}$ is measurable. Thus, $\liminf_{n\to\infty} f_n$ is also measurable.

□

Proposition 1.30: Stability under Passage to the Limit

Let $(E, \mathcal{A})$ be a measurable space and let $(f_n)_{n\in\mathbb{N}}$ be a sequence of $(\mathcal{A}, \mathcal{B}(\overline{\mathbb{R}}))$ - measurable functions with values in $\overline{\mathbb{R}}$. Then :

1. $\sup_n f_n$, $\inf_n f_n$, $\limsup_{n\to\infty} f_n$, and $\liminf_{n\to\infty} f_n$ are measurable.
2. If $(f_n)_{n\in\mathbb{N}}$ converges pointwise to a function f, then $f = \lim_{n\to\infty} f_n$ is measurable.
3. The convergence set $E_{\text{conv}} = \{x \in E \mid \lim_{n\to\infty} f_n(x) \text{ exists}\}$ is a measurable subset of E ($E_{\text{conv}} \in \mathcal{A}$).

Proof. We prove the measurability of the infimum and supremum operators:

§ **Measurability of** $f(x) = \inf_{n\in\mathbb{N}} f_n(x)$ **:** Using the sufficient condition for measurability, it suffices to prove that for every $a \in \mathbb{R}$, the preimage $f^{-1}([-\infty, a[)$ belongs to $\mathcal{A}$.

Observe that for any $x \in E$:

$$x \in f^{-1}([-\infty, a[) \iff f(x) < a \iff \inf_{n\in\mathbb{N}} f_n(x) < a$$

The infimum of a set of numbers is strictly strictly smaller than a if and only if at least one element of the sequence is strictly smaller than a:

$$\iff \exists n \in \mathbb{N}, \quad f_n(x) < a \iff x \in \bigcup_{n\in\mathbb{N}} \{x \in E \mid f_n(x) < a\}$$

Thus, we obtain the identity:

$$f^{-1}([-\infty, a[) = \bigcup_{n\in\mathbb{N}} f_n^{-1}([-\infty, a[)$$

Since each f_n is measurable, $f_n^{-1}([-\infty, a[) \in \mathcal{A}$ for every $n \in \mathbb{N}$. Being closed under countable unions, $\mathcal{A}$ contains this set, proving that $f = \inf_{n\in\mathbb{N}} f_n$ is measurable. □

§ **Measurability of** $g(x) = \sup_{n\in\mathbb{N}} f_n(x)$ **:** We proceed in an entirely analogous manner using intervals of the form $]a, +\infty]$:

$$g^{-1}(]a, +\infty]) = \left(\sup_{n\in\mathbb{N}} f_n\right)^{-1}(]a, +\infty]) = \bigcup_{n\in\mathbb{N}} f_n^{-1}(]a, +\infty]) \in \mathcal{A}$$

Hence, $g = \sup_{n\in\mathbb{N}} f_n$ is measurable. □

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